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Exp-7 Flow control at Data link layer

Aim:-
To write a Program to implement flow control at the Data link layer using the Sliding Window Protocol, and to Simulate the flow of frames from one node to another.

Algorithm:-
1. Start the Program.
2. Read the total no of frames to be sent (N), the Senders Window Size (W) and (optionally) a frame no to Simulate as lost during transmission.
3. Initialize base=0 and nextseq=0
4. Repeat while base < N:
   a. Determine the current Sender Window as the Range [base, base+W-1].
   b. Send all frames with Seq number from nextseq up to (base+W-1), incrementing nextseq after frame, Since the window restricts the Sender to at most W outstanding
   c. If the last frame lies within the frames just sent:
      (i) the receiver discards the lost frame and any subsequent out-of-order frames
      (ii) the Sender times out and retransmits all frames starting from the lost frame
   d. Else (no loss in this batch):

OUTPUT

=== SLIDING WINDOW PROTOCOL - Flow CONTROL SIMULATION (Go-Back-N) ===

Enter total no of frame : 8

Enter Sender's Window Size : 4

Enter frame no to simulate as lost (0 to 7, enter -1 for no loss) : 5

Sender window: [Frame 0 - Frame 3]

Node A → Sending frame 0

Node A → Sending frame 1

Node A → Sending Frame 2

Node B → Received Frame 0, Send Ack 0

Node B → Received Frame 1, Send Ack 1

Node B → Received Frame 2, Send Ack 2

Node B → Received Frame 3 Send Ack 3

Sender Window: [Frame 4 - Frame 7]

Node A → Send frame 5

Node A → Send frame 6

Node A → Send Frame 7

Node B → Received frame 4, Send Ack 4

Node B → Received Frame 5, Send Ack 5

Node B → Received Frame 6, Send Ack 6

Node B → Received frame 7, Send Ack 7

Node A → Ack 7 received Sliding Window forward

All 8 frames Successfully transmitted from Node A to Node B

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(2) The receiver accepts each frame in order and sends a cumulative acknowledgement (ack) for it.
(3) on receiving the ack for the last frame sent, the sender slides its window forward. base = last_seq + 1.
4. Repeat steps 2 and 3 until all N frames have been transmitted and acknowledged.
5. Stop the Program.

Program :-

```
def Simulate (total_frames, Window_Size, lost_frame):
    base = 0
    next_seq = 0
    while base < total_frames:
        window_end = min (base + Window_Size - 1, total_frames - 1)
        print ("\n ---------- ")
        print (f"Sender Window: [{base}, {window_end}]")

    while next_seq < base + Window_Size and next_seq < total_frames:
        print (f"Node A -> Sending frame {next_seq}")
        next_seq += 1

    if lost_frame is not None and base <= lost_frame < next_seq:
        print (f"Node B -> frame {lost_frame} missing. Discarding out-of-order frames after it")
```

```
        print(f"Node A -> Timeout! Resending
        Window Starting from frame {last_frame}")
        rent_seq = last_frame
        last_frame = None
        continue

    for i in range(base, rent_seq):
        print(f"Node B -> Received frame {i}
        Sending Ack {i}")
        print(f"Node A -> Ack {rent_seq - 1}
        received. Sliding window forward.")
        base = rent_seq
    print("\n----------")
print(f"All {total_frames} frames successfully
transmitted from Node A to Node B.")

def main():
    print("===SLIDING WINDOW PROTOCOL - Flow
    Control Simulation (Go-Back-N) ===\n")
    total_frames = int(input("Enter total no of
        frames to send: "))
    window_size = int(input("Enter sender window
        size: "))
    last_frame = int(input(f"Enter frame no
    to simulate as lost \n(0 to {total_frames-1},
    enter -1 for no loss): "))
    if last_frame == -1:
        last_frame = None
    simulate(total_frames, window_size, last_frame)

if __name__ == "__main__":
    main()
```

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Result :-

The Program to Implement flow Control at the Data Link layer using the Sliding Window Protocol was Successfully written in Python, executed and Verified by Simulating the flow of frames b/w two Nodes.